

■ ENTERPRISE SUPPORT PLATFORM

Modern Full-Stack Helpdesk Platform

Sub-Millisecond Query Budgets • PostgreSQL 17 • Go 1.23 • Single-Page App

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Core Stack:
Go 1.23, PostgreSQL 17, Redis 7

Architecture:
Transactional Outbox & Keyset SPA

Open Source Repository:
github.com/osamazakimohammed/helpdesk

Executive Summary & Core Value Proposition

High-performance support platform built for sub-millisecond responsiveness and simplicity.

■ Zero-Bloat Sub-Millisecond Speed

Eliminates heavy JS framework dependencies and slow template rendering. Strict query budget middleware enforces sub-millisecond database queries across all endpoints.

■ Uncompromising Transactional Safety

Employs the **Transactional Outbox Pattern** with atomic PostgreSQL commits. Ticket updates, audit records, and async side-effects never desynchronize or fail silently.

■ Universal Multilingual & RTL Ready

Native UTF-8 support with automatic bidirectional text rendering (dir='auto'). Seamless right-to-left alignment for Arabic subjects, descriptions, and comments.

■ Single-Binary Deployment (Self-Hostable)

The entire Single-Page Application (HTML, CSS, JS) is embedded directly into the static Go binary using embed.FS. One container image runs the entire system.

System Architecture & Technology Stack

Engineered with compiled type safety, robust connection pooling, and minimal footprint.

■ Backend Core (Go 1.23)

- **Chi Router:** Ultra-lightweight, zero-allocation HTTP router.
- **pgx/v5 Pool:** High-throughput connection pooling with prepared statements.
- **SQLC:** Compile-time type-safe query generation eliminating ORM overhead.
- **JWT & Middleware:** Query budget enforcer, CORS, structured logging.

■ Database & Storage (PostgreSQL 17)

- **UUIDv7:** Time-ordered 128-bit UUID primary keys for natural indexing.
- **BRIN Indexes:** Ultra-compact range index on high-volume audit logs.
- **Keyset Pagination:** Pure (updated_at, id) cursor ordering without OFFSET.
- **S3 Object Storage:** MinIO/AWS S3 compatibility with SHA-256 validation.

■ Caching & Async Workers (Redis 7)

- **Distributed Outbox Drainer:** SKIP LOCKED concurrent queue.
- **Rate Limiting:** Token bucket protection on authentication and intake.
- **PubSub:** Real-time event broadcasting and timeline synchronization.
- **Idempotency:** Guaranteed deduplication on retry loops.

■ Embedded Frontend (Vanilla SPA)

- **Zero-Dependency SPA:** Pure vanilla JS + reactive DOM rendering.
- **Live Hot-Reload:** Development mode serves disk assets instantly.
- **Distroless Shipping:** Static single binary container ready for Podman/Docker.
- **Auto RTL Alignment:** Native dir='auto' for global readability.

Dedicated Application Surfaces & Workflows

Unified experiences tailored for Support Agents, Customers, and Public Users.

■ Agent Workspace (/app)

- **Unified Queues:** All Open, Pending Response, Resolved / Closed, All Tickets.
- **Status Transitions:** Real-time dropdown updates with database timestamp recording.
- **Dual-Mode Composer:** Send Public Customer Replies or Save Internal Team Notes.
- **Keyboard Shortcuts:** Fast triage with *Ctrl+Enter* submission.

■ Customer Portal (/portal)

- **Self-Service Overview:** Instant list of all submitted tickets and live statuses.
- **Threaded History:** Real-time conversation stream with agent response badges.
- **Interactive Replies:** Customers reply directly into existing ticket threads.
- **Arabic & Multilingual:** Clean Right-to-Left aligned text display.

■ Request Intake Form (/submit)

- **Streamlined Intake:** Quick ticket creation for authenticated or guest users.
- **Priority & Categories:** Select issue urgency (Low, Medium, High, Urgent).
- **Auto Identity Binding:** Resolves logged-in contact IDs automatically.
- **Immediate Redirection:** Seamless jump to ticket view upon submission.

■ Knowledge Base & API Docs (/kb & /api/docs)

- **Knowledge Base (/kb):** Public spaces, categories, and fast full-text search.
- **Interactive Swagger UI (/api/docs):** Live OpenAPI 3.1 schema exploration.
- **Universal Access:** Accessible for team members, clients, and developers alike.

Performance Engineering & Strict Query Budgeting

Eliminating the N+1 problem through compile-time SQL aggregation and keyset pagination.

■ Strict Query Budgets

- **Ticket Queue List: STRICTLY 1 Query**

CTE & JSON subquery aggregation bundles tags, assignee, contact, status, and organization in a single database round-trip.

- **Ticket Detail Pane: STRICTLY 2 Queries**

Query 1 fetches ticket metadata; Query 2 retrieves keyset-paginated timeline events.

■ Keyset Pagination (No OFFSET)

- **Constant-Time Retrieval:** Uses WHERE (t.updated_at, t.id) < (\$cursor_time, \$cursor_id).

- **Zero Scan Penalty:** Page 1,000 loads with the exact same 0.4ms index-seek latency as Page 1.

- **Drift-Free:** New incoming tickets do not cause duplicate items during pagination.

■ Middleware Enforcement

- **Query Budget Middleware:** Tracks total database queries per HTTP request.

- **Header Transparency:** Emits X-DB-Query-Count in response headers for latency & efficiency verification.

- **Circuit Breaker:** Flags requests exceeding the query limit.

■ UUIDv7 & BRIN Indexing

- **UUIDv7 Primary Keys:** Time-ordered 128-bit UUIDs prevent B-Tree fragmentation while eliminating sequential enumeration vulnerabilities.

- **BRIN Index on Audit Trail:** Compresses millions of immutable audit logs into tiny page-range metadata for blazing fast chronological queries.

Authentication, Roles & Internationalization

Frictionless login, universal admin privileges, and first-class Arabic support.

■ Dual Username / Email Login

- Users can authenticate using **either their plain username** (e.g. admin, support, customer) **or their full email address**.
- Returns standard cryptographically signed JWT tokens with claims.
- Automated session persistence via secure browser storage.

■ Universal Admin Access

- **Admin Role:** Direct navigation across *every surface* (Agent Workspace, Customer Portal, Submit Request, Knowledge Base, OpenAPI).
- **Agent Role:** Focused ticket resolution workspace & Help Center.
- **Customer Role:** Personal portal and ticket submission.

■ Native Arabic & RTL Text Direction

- Comprehensive UTF-8 encoding across PostgreSQL, Go, and Browser.
- Automatic dir='auto' attribute dynamically detects Arabic script and renders right-to-left alignment flawlessly.
- Tested with full Arabic subjects, ticket descriptions, and agent replies.

■ Demo Accounts & Seeding

- **Admin:** admin / admin (Full platform access)
- **Support:** support / support (Agent workspace)
- **Customer:** customer / customer (Portal & intake)
- Clean seed migrations populate standard priorities and statuses.

Quickstart, Deployment & Live Repository

Ready to run locally with Podman/Docker or deploy as a single static binary.

■ 1-Click Container Startup

Start PostgreSQL 17 & Redis 7

```
podman run -d --name helpdesk-postgres -p 5432:5432 -e POSTGRES_USER=postgres -e POSTGRES_PASSWORD=postgres -e POSTGRES_DB=helpdesk postgres:17-alpine
```

Or using Docker Compose

```
docker compose up -d
```

■ CLI Commands & Server Launch

1. Run database schema migrations

```
go run cmd/server/main.go migrate
```

2. Seed default roles & demo accounts

```
go run cmd/server/main.go seed
```

3. Start live server on :8080

```
go run cmd/server/main.go serve
```

■ Cloud Deployment Options

- **Render / Railway:** 1-click Git deployment from repository.
- **Fly.io:** Global static distroless container deployment.
- **Self-Hosted VPS:** Run on any Linux server with Docker Compose.
- **Distroless Base:** Google distroless non-root security container.

■ Repository & Documentation

• GitHub Repository:

<https://github.com/osamazakimohammed/helpdesk>

• **Live Local Server:** <http://localhost:8080>

• **OpenAPI 3.1 Swagger Docs:** <http://localhost:8080/api/docs>